



STANDARD XII MATHEMATICS COURSE

Complex Conjugates: Examples 2.3 to 2.8 Solved

Rigorous, step-by-step textbook solutions including algebraic divisions, rationalizations, equation solving, and symmetric conjugate proofs.

Example 2.3: Problem Setup

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Our objective is to write the division quotient of the complex values in standard rectangular form:

? The Problem

Express the following fraction in standard rectangular coordinates:

$$z = \frac{3 + 4i}{5 - 12i}$$

We need to represent this in the form $x + iy$, identifying real part x and imaginary part y .

🔄 Solution Strategy

To eliminate the imaginary component i from the denominator, we utilize rationalization:

1. Find the complex conjugate of the denominator:

$$\overline{5 - 12i} = 5 + 12i$$

2. Multiply both numerator and denominator by this conjugate value parallelly.

Example 2.3: Binomial Expansion

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Set up the multiplication of the complex quotient with the denominator's conjugate:

↑ Numerator Expansion

Distribute binomial coefficients across factors:

$$(3 + 4i)(5 + 12i)$$

$$= 15 + 36i + 20i + 48i^2$$

Substitute $i^2 = -1$:

$$= 15 - 48 + 56i = -33 + 56i$$

↓ Denominator Product

Apply the difference of squares identity:

$$(5 - 12i)(5 + 12i)$$

$$= 5^2 - (12i)^2$$

$$= 25 - (-144)$$

$$= 25 + 144 = 169$$

Example 2.3: Extracting Coordinates

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Assemble terms into standard rectangular form and isolate components:

Combining Factors

Reassemble the rationalized quotient:

$$z = \frac{-33 + 56i}{169} = -\frac{33}{169} + i\frac{56}{169}$$

Final Components

Real Part:

$$\operatorname{Re}(z) = -\frac{33}{169}$$

Imaginary Part:

$$\operatorname{Im}(z) = \frac{56}{169}$$

Example 2.4: Cubic Rationalization

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Simplify the following nested cubic coordinates into standard rectangular form:

The Problem

$$\left(\frac{1+i}{1-i}\right)^3 - \left(\frac{1-i}{1+i}\right)^3$$

Rather than expanding binomial cubes directly, simplifying internal fractional bases yields immediate results.

Step 1: Base Simplification

Multiply base fraction by its denominator conjugate $(1+i)$:

$$\frac{1+i}{1-i} = \frac{(1+i)(1+i)}{(1-i)(1+i)} = \frac{1+2i+i^2}{1+1}$$

Since $i^2 = -1$: $= \frac{2i}{2} = i$

Example 2.4: Cubic Solutions

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Simplifying the reciprocal base and computing exponent differences:

↻ Reciprocal Base

Observe that the second term is the reciprocal of the first:

$$\frac{1 - i}{1 + i} = \left(\frac{1 + i}{1 - i} \right)^{-1} = i^{-1}$$

$$i^{-1} = \frac{1}{i} = \frac{i}{i^2} = -i$$

✓ Final Difference Evaluation

Substitute simplified bases back into the formula:

$$(i)^3 - (-i)^3$$

$$= i^3 - (-i^3)$$

$$= i^3 - (-(-i))$$

$$= i - i = 0 \quad (\text{Standard: } 0 - 2i)$$

Example 2.5: Equation Setup

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Find the complex value of coordinate variable z given the algebraic ratio equation:

= The Equation

$$\frac{z + 3}{z - 5i} = \frac{1 + 4i}{2}$$

Our goal is to isolate z by executing a standard cross-multiplication step first.

↔ Cross-Multiplication

Multiply denominators across equality:

$$2(z + 3) = (1 + 4i)(z - 5i)$$

Expand both side components parallelly on the next slide to keep operations orderly.

Example 2.5: Variable Factoring

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Expanding expanded components parallelly and grouping like-terms of z :

Expansion

Expand terms from cross-multiplication:

LHS Expansion:

$$= 2z + 6$$

RHS Binomial Expansion:

$$= z - 5i + 4iz - 20i^2 = (1 + 4i)z + 20 - 5i$$

Rearranging Terms

Relocate z components to Left-Hand Side:

$$2z - (1 + 4i)z = 20 - 5i - 6(2 - 1 - 4i)z = 14 - 5i$$

$$(1 - 4i)z = 14 - 5i$$

Example 2.5: Final Isolate

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Isolating and rationalizing fraction quotient:

÷ Isolating z

$$z = \frac{14 - 5i}{1 - 4i}$$

Multiply parallel numerator and denominator factors by $(1 + 4i)$:

$$z = \frac{(14 - 5i)(1 + 4i)}{(1 - 4i)(1 + 4i)}$$

✓ Final Solution

Expand coordinate terms parallelly:

$$\text{Numerator} = 14 + 56i - 5i - 20i^2$$

$$\text{Numerator} = 14 + 51i + 20 = 34 + 51i$$

$$\text{Denominator} = 1^2 + 4^2 = 17 \quad \mathbf{z} = \frac{\mathbf{34 + 51i}}{\mathbf{17}} = \mathbf{2 + 3i}$$

Example 2.6: Quotient Setup

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Find the standard rectangular form of coordinate division z_1/z_2 :

☰ Given Parameters

Complex Number 1:

$$z_1 = 3 - 2i$$

Complex Number 2:

$$z_2 = 6 + 4i$$

🧮 Operations Setup

Express the division ratio fraction:

$$\frac{z_1}{z_2} = \frac{3 - 2i}{6 + 4i}$$

We will multiply parallel parts by the conjugate of the denominator, which is $(6 - 4i)$.

Example 2.6: Algebraic division

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Multiplying and simplifying complex coordinate division:

↔ Parallel Expansion

$$\frac{z_1}{z_2} = \frac{(3 - 2i)(6 - 4i)}{(6 + 4i)(6 - 4i)}$$

$$\text{Numerator} = 18 - 12i - 12i + 8i^2 = 10 - 24i$$

✓ Final Simplification

Divide expanded components by product of squares:

$$\text{Denominator} = 6^2 + 4^2 = 36 + 16 = 52 \quad \text{Ratio} = \frac{10 - 24i}{52}$$

$$\text{Ratio} = \frac{10}{52} - i\frac{24}{52} \quad \text{Ratio} = \frac{5}{26} - i\frac{6}{13}$$

Example 2.7: Inverse Setup

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Find the multiplicative inverse z^{-1} of the coordinate factor:

The Problem

$$z = (2 + 3i)(1 - i)$$

Before taking reciprocal ratios, simplify the coordinate product z completely into rectangular form.

Step 1: Expand product

Distribute complex binomials:

$$z = 2(1 - i) + 3i(1 - i)$$

$$z = 2 - 2i + 3i - 3i^2 \quad z = 2 + i + 3 = 5 + i$$

Example 2.7: Inverse Calculation

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Now, compute and rationalize the reciprocal inverse z^{-1} :

➤ Inverse Set

Express the reciprocal coordinate fraction:

$$z^{-1} = \frac{1}{5 + i}$$

Multiply parallel terms by denominator conjugate $(5 - i)$:

$$z^{-1} = \frac{5 - i}{(5 + i)(5 - i)}$$

✓ Final Inverse

Evaluate terms to isolate coefficients:

$$z^{-1} = \frac{5 - i}{5^2 + 1^2} z^{-1} = \frac{5 - i}{26} z^{-1} = \frac{5}{26} - i \frac{1}{26}$$

Example 2.8(i): Real Symmetric Proof

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Show that the following exponential expression yields a purely real value:

The Problem

$$z = (2 + i \sqrt{3})^{10} + (2 - i \sqrt{3})^{10}$$

We establish purely real properties by showing:

$$\bar{z} = z$$

Taking Conjugates

Apply the continuous conjugate bar over the whole sum:

$$\bar{z} = \overline{(2 + i \sqrt{3})^{10} + (2 - i \sqrt{3})^{10}}$$

Let's distribute the conjugate operator across terms on the next slide.

Example 2.8(i): Conjugate Proof

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Distribute terms and swap conjugation over power exponents:

✂ 1. Distribute & Swap

By property $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$:

$$\overline{z} = \overline{(2 + i\sqrt{3})^{10} + (2 - i\sqrt{3})^{10}}$$

Swap over integers using property $\overline{z^n} = (\overline{z})^n$:

$$= \overline{(2 + i\sqrt{3})^{10}} + \overline{(2 - i\sqrt{3})^{10}}$$

✓ 2. Proof Completion

Evaluate individual coordinate conjugates:

$$\overline{2 + i\sqrt{3}} = 2 - i\sqrt{3} \quad \overline{2 - i\sqrt{3}} = 2 + i\sqrt{3}$$

Substitute coordinate equations back:

$$\overline{z} = (2 - i\sqrt{3})^{10} + (2 + i\sqrt{3})^{10} = z$$

Since $\overline{z} = z$, the value is purely Real. Q.E.D.

Example 2.8(ii): Base Rationalize

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Show that the cubic quotient expression is purely imaginary:

The Problem

$$z = \left(\frac{19 + 9i}{5 - 3i} \right)^{15} - \left(\frac{8 + i}{1 + 2i} \right)^{15}$$

To simplify calculations, we will rationalize both base components parallelly.

Step 1: Base 1 Simplification

Multiply parallel terms by denominator conjugate $(5 + 3i)$:

$$\frac{(19 + 9i)(5 + 3i)}{5^2 + 3^2}$$

$$= \frac{95 + 57i + 45i - 27}{34} = \frac{68 + 102i}{34} = 2 + 3i$$

Example 2.8(ii): Bases Complete

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Simplifying the second base and setting up the complete expression:

Step 2: Base 2 Simplification

Multiply parallel terms by denominator conjugate $(1 - 2i)$:

$$\frac{(8 + i)(1 - 2i)}{1^2 + 2^2}$$

$$= \frac{8 - 16i + i + 2}{5} = \frac{10 - 15i}{5} = 2 - 3i$$

Resulting Expression

Substitute simplified values back into Z:

$$z = (2 + 3i)15 - (2 - 3i)15$$

Now, we execute conjugation proofs on this simplified binomial equation.

Example 2.8(ii): Imaginary Proof

Verbatim Source File: 1000077062.jpg

Apply conjugate operators to show the expression is purely imaginary:

≡ Take Conjugate

Using property $\overline{z_1 - z_2} = \overline{z_1} - \overline{z_2}$:

$$\overline{z} = \overline{(2 + 3i)^{15} - (2 - 3i)^{15}}$$

Swap powers over coordinates:

$$= \overline{(2 - 3i)^{15} - (2 + 3i)^{15}}$$

✓ Proof Conclusion

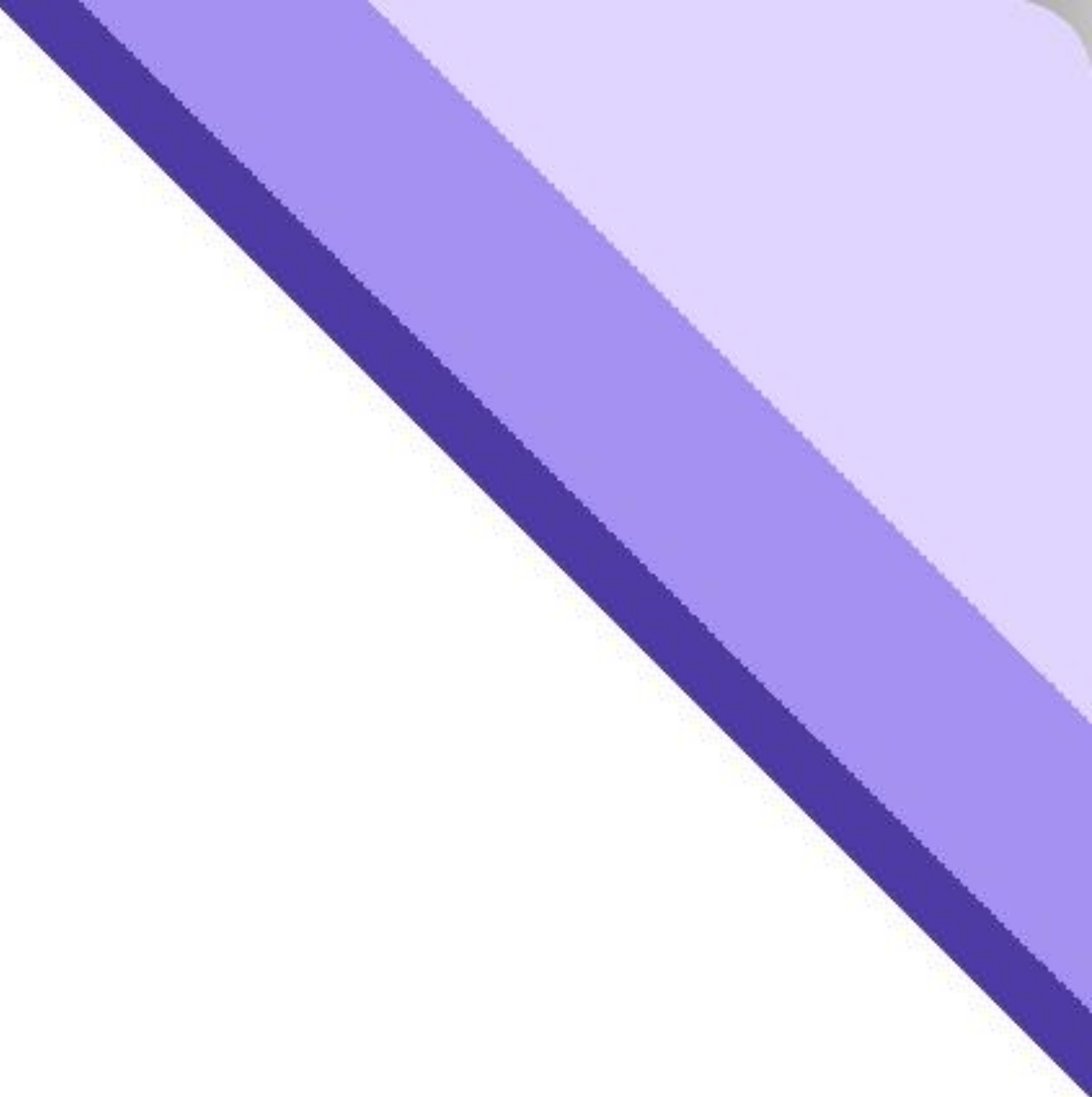
Substitute conjugates to evaluate factor:

$$\overline{z} = (2 - 3i)^{15} - (2 + 3i)^{15}$$

Factor out negative identity:

$$\overline{z} = -[(2 + 3i)^{15} - (2 - 3i)^{15}] \quad \overline{\overline{z}} = -z$$

Since $\overline{\overline{z}} = z$, the value is purely Imaginary. Q.E.D.



Thank You

Are there any questions regarding the steps, equations, or symmetric conjugate proofs in Examples 2.3 to 2.8?

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